

Caffeine intake and atrial fibrillation incidence: dose response meta-analysis of prospective cohort studies.

BACKGROUND: The association between habitual caffeine intake with incident atrial fibrillation (AF) was unknown. We conducted a meta-analysis to investigate the association between chronic exposure of caffeine and the risk of AF and to evaluate the potential dose-response relation. **METHODS:** We searched PubMed, EMBASE, and the Cochrane Library up to November 2013 and references of relevant retrieved articles. Prospective cohort studies were included with relative risk (RR) or hazard ratio and 95% confidence intervals (CIs) for AF according to coffee/caffeine intake. **RESULTS:** Six prospective cohort studies with 228,465 participants were included. In the primary meta-analysis, caffeine exposure was weakly associated with a reduced risk of AF (RR, 0.90; 95% CI, 0.81-1.01; $P = 0.07$; $I(2) = 73\%$). In subgroup analyses, pooled results from studies with adjustment of potential confounders showed an 11% reduction for low doses (RR, 0.89; 95% CI, 0.80-0.99, $P = 0.032$; $I(2) = 30.9\%$, $P = 0.227$) and 16% for high doses (RR, 0.84; 95% CI, 0.75-0.94, $P = 0.002$; $I(2) = 24.1\%$, $P = 0.267$) of caffeine consumption in AF risk. An inverse relation was found between habitual caffeine intake and AF risk (P for overall trend = 0.015; P for nonlinearity = 0.27) in dose-response meta-analysis and the incidence of AF decreased by 6% (RR, 0.94; 95% CI, 0.90-0.99) for every 300 mg/d increment in habitual caffeine intake. **CONCLUSIONS:** It is unlikely that caffeine consumption causes or contributes to AF. Habitual caffeine consumption might reduce AF risk. Copyright © 2014 Canadian Cardiovascular Society. Published by Elsevier Inc. All rights reserved.